

AeroGen X1 Cooling System — Operations Manual

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The normal operating temperature of the avionics cooling loop is 60–75 °C. A warning is raised at 85 °C. A critical shutdown is required at 95 °C or above. Pump P-17 is the primary circulation pump. If a thermal warning is accompanied by elevated pump vibration, inspect the P-17 bearings and verify airflow through cooling manifold CM-4.

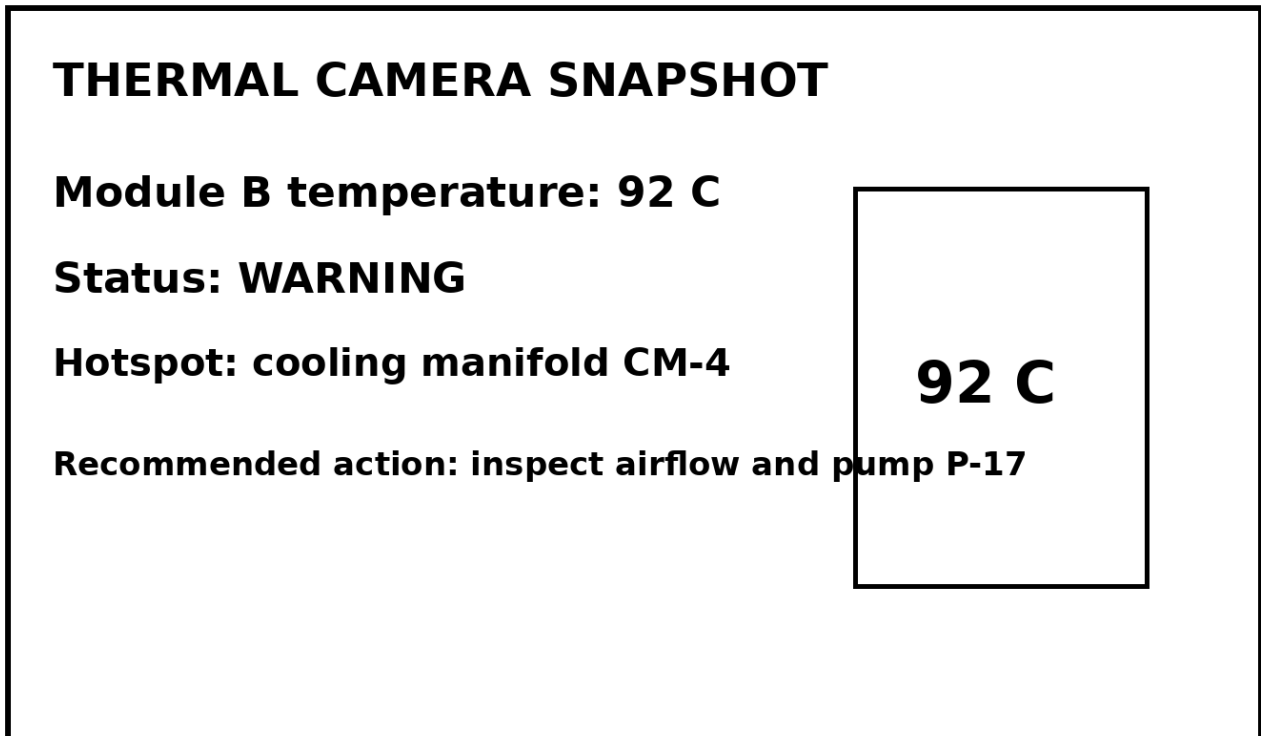
Component limits

Component	Normal range	Warning / Limit	Action
Pump P-17 vibration	0–3.0 mm/s	4.5 mm/s max	Inspect bearings above limit
Loop temperature	60–75 °C	85 °C warning	Inspect cooling path
Coolant pressure	2.1–2.8 bar	1.8 bar minimum	Check for leak / pump

Maintenance note: after replacing P-17 bearings, perform a 10-minute run and confirm vibration remains below 3.0 mm/s.

Appendix A — Thermal inspection evidence

The following image is a field thermal-camera snapshot captured during troubleshooting. Information visible inside the image is intentionally not repeated as normal PDF text; the interview solution should extract image information separately.



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